

SEDAR 5: Atlantic and Gulf of Mexico King Mackerel Assessment Workshop

Assessment Data

- Available growth data needs to be evaluated for improved application to historical catch at age.
- Available sex ratio at size data needs to be evaluated to determine how sex ratios vary by size.
- Methods that allow for including error estimates in the catch at age matrix need to be developed.
- Continued evaluation of tag data, ongoing otolith microchemistry and shape analysis studies, and micro-satellite genetic marker data to improve estimation of stock structure and mixing proportions.
- Field studies are needed to develop or improve batch fecundity, spawning frequency, and age specific fecundity estimates, including size and age at maturity.
- Western Gulf king mackerel catches need to be aged for use in age length key analyses.

Assessment Modeling

- Currently, it is only possible to model two stocks using tagging data to model mixing rates (Porch 2003). In the long term the Data Workshop and Assessment Panels recommend that assessment models be developed which can model multiple stocks and/or areas and which can use multiple types of data that enable mixing rate estimations (including tagging data and biological tags including elemental and isotopic composition, genetic information and morphological information).
- Sensitivity of CAA and management benchmarks to changes in the growth model used in the stochastic ageing procedure need to be evaluated.
- A three-area age structured model with forward projection formulation may result in better estimation of the impact on stock status of mixing zone dynamics using existing tagging data and most recent recruitment estimates.
- Sensitivity runs considered in this assessment indicate two areas where additional research is critically needed to improve stock status evaluation. The Assessment Workshop Panel advises that stock assessment uncertainty will not be reduced until these issues are resolved. These two areas are:

- Methods used to allocate catches to age class when samples are inadequate for constructing age-length keys. Sensitivity runs based on alternative growth models suggest that estimates of stock status are sensitive to differences in growth models when they are used to estimate age from size in the absence of an ALK. The raw data used to develop the historical growth models (Manooch et al. 1987; Collins et al. 1988) are no longer available, and thus it may not be possible to provide the type of identical analyses of current and historic data that are necessary to evaluate whether growth model differences are simply due to analytical technique or whether the differences truly reflect changes in growth over time. The Panel recommends that current growth data (1987 onward) be modeled with increased resolution to refine growth model parameters. Specifically, decimal rather than integer ages should be modeled, and attention should be paid to collection date, birth date, and annulus formation date.
- Sensitivity analyses of stock mixing impacts on stock status determination. Results suggest that the assumed degree of stock mixing has relatively equivalent impacts on the perceived productivity of each migratory units, but divergent impacts on stock status determination. The estimated status of the Gulf of Mexico Migratory Unit is strongly influenced by mixing assumptions, while status determination of the Atlantic Migratory Unit varies minimally. Both the Data and Assessment Workshop Panels devoted significant discussion and effort toward resolving stock allocation within the mixing zone. Based on Data Workshop recommendations, the SEFSC reconsidered mixing rates through updated analyses of tag data, developed an alternative assessment framework to incorporate tag-based stock mixing estimates into a VPA framework, and developed stock estimates with the base assessment configuration for a variety of mixing rates within the mixing zone. However, none of these efforts have led to a consensus recommendation on the actual level of stock mixing.
- The Assessment Workshop Panel believes that analyses of otolith shape and microchemistry, as presented in the progress reports discussed at the Data Workshop, offer a promising approach to resolving stock mixing. The Assessment Workshop Panel strongly recommends that this work be continued for several additional years to increase sample size, continually improve the resolution of the method, and better account for potential annual variation in mixing. The Panel also recommends increased sampling intensity within the mixing zone, with sample allocation that is representative of the fine-scale geographic distribution of the catch within the mixing zone. Also an effective tagging program designed specifically to address the mixing issue could increase the quality and quantity of available data.

Review Workshop

Research Recommendations

- The RW Panel noted that major concerns remain about the growth curves used to age the catch in some years and areas, the fecundity-length relationship used to estimate spawning stock, and the degree of mixing of the Gulf and Atlantic migratory groups in the winter fishery mixing zone. The RW Panel also expressed concern about the limited number of fishery independent indices of abundance available for VPA calibration.
- The RW Panel recommends enhancing ongoing research programs and implementing new research programs to collect fishery independent data (e.g., length measurements, age structures, fecundity measurements) to improve the accuracy and precision of current estimates of growth, fecundity, and stock mixing. Spatial variability in size at maturity and fecundity at age should be evaluated among regions/migratory groups.
- The data collection program should also be designed to provide fisheries independent indices of abundance for the full age range in the stock. This consideration should have a strong influence on the design aspects [gear, season] of the recommended research programs. These programs might include research sampling targeting spawning aggregations, research sampling targeting juveniles, tagging studies specifically designed to provide information on mixing rates, and hydro-acoustic sampling. Scientists should seek the advice of members of the commercial and recreational fishing communities in the design of these programs.
- The RW Panel suggested that the MRFSS indices of abundance could be recompiled to address two issues: 1) consider incorporation of the January-June intercept data in addition to the current July-December data, and 2) consider restriction of the sample data to the age classes most likely to contribute to the respective catch types (i.e., recompile the indices including only Catch Types A, and restrict the corresponding length composition to legally landed fish).
- The RW Panel also recommended the future application of different assessment models to provide alternative perspectives on the status of king mackerel stocks (e.g., those including estimation of the likely degree of error in the fishery catch-at-age, and/or those which employ forward-projecting computation approaches).
- One growth model should be developed for the splitting of catch at length data into catch at age data and another one that can be used for stock related data like weight at age in the stock, maturity at age in the stock and the like.
- Available sex ratio at size data needs to be evaluated to determine how sex ratios vary by size.
- Western Gulf king mackerel catches need to be aged for use in age length key analysis.

- The commercial fishery tuning indices should be further developed and it seems important that this is done in cooperation with fishers with an intimate knowledge of the way the fishery is prosecuted.
- Age composition of commercial and recreational discards is needed.
- Discard mortality rates are needed.
- Tuning indices should be weighted according to their internal variability, the part of the stock covered by the index, correlation with other indices etc. For instance it is realized that using their individual degree of correlation to the VPA stock abundance estimates could be problematic due to the circular logic feature of this approach.
- Data from Mexican catches need to be obtained, probably via initiatives for closer cooperation with Mexico. In this connection there is a need to look into whether the eastern and western Gulf King Mackerel are separate stock components.
- Tagging programs specifically designed to examine the mixing should be developed. Otolith shape and microchemistry and maybe micro-satellite DNA analysis are promising methods that should be pursued.
- Mixing of the stocks in the mixing zone should be investigated also the during summer period.

Process and Procedure Recommendations

- The amount of documentation and issues to be dealt with are significant. Some of the documentation could have been sent out earlier to the RW Panel, for instance background material and the data workshop material. That would have eased the task of getting deeply into the substance of the material, especially for the external reviewers, who (almost by definition) were not beforehand familiar with the assessment.

Comments from the CIE Contractors

(These are excerpted comments intended to highlight suggestions and areas of concern; readers are encouraged to consult the full report for additional details)

- The amount of reports and other material to read before the meeting was extensive. There was only little time to do this, about two weeks. It would be useful if some of the material were sent out as early as possible. It should be possible to send out previous assessment reports, background articles, and the Data Workshop report, several weeks earlier.
- A complete description of the assessment with all the input data files and the precise settings of the model would be nice to have in one document. It was a bit difficult to find

precisely in which document to look for the various details. The level of details and data files should allow for an exact and easy repeat of the calculations.

- Fishers (and nongovernmental organizations (NGOs)) contributed during the meeting some information on CPUE series, the fishery and the management regulations effects on this, and the like. It is, however, important that political issues do not enter the discussions. It might, however, be important for the entire process that fishers participate, or at least get the opportunity to observe what is done, in order to secure transparency and trust in the system. However, extra time would need to be spent on explaining things to non-scientists and in balancing the statements put forward so that fishers and NGOs correctly understand the issues.
- My task as Chair for the meeting was a bit difficult because most panel members were more familiar with the process than I was. Maybe a bit more information about the duties of the Chair would be useful. Alternatively, another member of the Panel could be the Chair, and one of the CIE Experts could be appointed as the lead expert and perhaps still be responsible for the reporting.
- Maybe the reviewers (and other Panel members) could, to the extent possible, state before the start of the meeting what sensitivity runs they want to see in addition to what has been presented in the documents sent to the Panel. This will allow SEFSC staff more time to prepare the runs, and it will make mistakes less likely.
- The timing of the whole process from the last data sampled in 2001/02 and until now (start of 2004) with the aim of giving advice for 2005/2006 could be improved. It is a very long time span, and there is a large risk for the things in the fishery and the stock to have changed in between meeting processes. It should be possible to shorten this time span so that the advice for 2005/2006 can be based on data from 2003/2004.
- It is important that estimates of age-composition of commercial and recreational discards, and of discard mortality be obtained. It is strongly recommended that fisheries-independent surveys be expanded, and eventually assigned more weight in the tuning process.
- Fisheries-independent surveys should be designed to provide indices of abundance for the full age range in the stock. This would likely require multi-seasonal sampling and the combined use of multiple sampling gears and hydro-acoustics.
- Data from Mexican catches need to be obtained to improve the accuracy of Gulf king mackerel assessments.
- If feasible, I recommend that the uncertainty in assessments caused by sampling variability in estimated landings in number by age be further evaluated.

- The use of multiple survey indices for “tuning” can introduce a bias of unknown magnitude in the assessments of Atlantic and Gulf king mackerel. One way to reduce such bias is to combine overlapping survey estimates by using a composite estimator with weights determined by coverage and precision of each abundance series, and then apply the combined series in tuning the model. Additional post-stratification might be appropriate when surveys overlap only in a sub-area or during a limited time.